

GALUUT, Mongolia

WMO: 442840

Lat: 46.7020N

Lon: 100.1430E

Elev: 6975

StdP: 11.35

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-29.6	-39.1	0.8	-31.5	-35.1	1.0	-27.7	17.4	2.6	13.4	-1.0	1.3	0	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	77.4	56.5	74.2	54.6	71.4	53.2	60.3	71.7	58.0	69.1	55.9	66.8	7.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.3	87.8	65.0	53.7	79.8	63.1	51.4	73.1	60.6	30.2	71.7	28.3	69.2	26.7	66.8	72.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	17.5	14.7	-37.1	83.0	5.1	2.9	-40.7	85.1	-43.7	86.8	-46.6	88.4	-50.3	90.6		
			-37.2	62.5	5.1	5.0	-40.8	66.1	-43.8	69.0	-46.7	71.8	-50.4	75.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-12.4	-5.0	13.5	33.0	44.6	55.2	59.7	55.9	44.0	27.8	8.0	-7.1	(6)
(7)		DBStd	26.59	8.42	9.90	11.88	9.01	7.73	6.18	5.40	5.33	7.55	8.68	10.99	8.49	(7)
(8)		HDD50	9276	1936	1540	1130	510	200	23	2	12	202	690	1261	1770	(8)
(9)		HDD65	14043	2401	1960	1595	959	633	299	180	286	629	1155	1711	2235	(9)
(10)		CDD50	732	0	0	0	1	32	179	302	195	23	0	0	0	(10)
(11)		CDD65	23	0	0	0	0	0	4	15	4	0	0	0	0	(11)
(12)		CDH74	289	0	0	0	0	3	65	166	55	0	0	0	0	(12)
(13)		CDH80	22	0	0	0	0	0	4	14	4	0	0	0	0	(13)
(14)	Wind	WSAvg	3.7	1.4	1.9	4.0	6.1	6.6	5.2	4.4	3.7	3.8	3.4	2.3	1.7	(14)
(15)	Precipitation	PrecAvg	8.3	0.1	0.1	0.2	0.3	0.6	1.5	2.5	2.2	0.7	0.3	0.1	0.1	(15)
(16)		PrecMax	18.1	0.4	0.3	0.7	1.9	4.4	5.4	4.7	5.6	2.1	1.5	0.6	0.4	(16)
(17)		PrecMin	3.3	0.0	0.0	0.0	0.0	0.0	0.1	0.7	0.3	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.7	0.1	0.1	0.2	0.4	0.7	1.1	1.1	1.2	0.5	0.3	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.5	26.3	47.5	62.5	73.5	80.0	82.1	79.5	70.0	56.6	41.2	20.8	(19)
MCWB			11.4	20.8	33.4	45.8	50.2	57.9	59.5	57.2	49.7	40.5	32.3	16.5	(20)	
2%		DB	9.6	19.6	41.9	57.7	68.7	75.3	78.5	74.6	65.9	51.8	34.8	14.0	(21)	
		MCWB	7.0	16.2	29.7	41.0	48.3	55.1	57.5	54.2	48.0	39.1	27.9	11.0	(22)	
5%		DB	4.7	14.7	37.1	53.6	64.6	72.2	75.3	71.3	62.3	47.5	29.5	9.9	(23)	
		MCWB	3.1	12.1	27.5	38.4	46.2	53.0	56.0	53.3	46.4	36.0	24.3	7.7	(24)	
10%		DB	0.4	10.5	32.2	49.5	60.8	69.0	72.1	68.2	58.6	43.1	24.9	6.0	(25)	
		MCWB	-0.8	8.5	25.1	36.3	43.9	51.6	54.6	52.0	44.5	33.5	20.6	4.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	21.4	34.3	48.6	54.4	62.0	64.2	61.9	52.7	43.2	32.8	16.8	(27)
MCD B			14.9	25.1	44.9	59.4	68.9	75.8	75.1	72.9	64.4	53.2	40.3	20.3	(28)	
2%		WB	7.0	16.2	30.6	42.9	50.7	58.6	61.3	58.1	50.0	39.5	28.3	11.4	(29)	
		MCD B	9.3	19.4	39.9	55.0	64.1	71.0	72.8	67.4	61.7	49.9	34.5	13.9	(30)	
5%		WB	3.2	12.2	28.0	39.5	48.1	55.8	59.1	55.8	47.8	36.8	24.2	7.9	(31)	
		MCD B	4.7	14.7	36.5	51.6	61.5	66.9	70.6	66.1	60.1	46.4	29.4	9.9	(32)	
10%		WB	-0.8	8.6	25.2	36.7	45.5	53.8	56.9	53.9	45.8	33.8	20.8	4.5	(33)	
		MCD B	0.3	10.6	32.2	48.5	58.5	64.8	68.3	64.9	56.6	42.4	24.9	6.0	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.9	23.9	25.6	22.8	22.9	21.9	20.3	21.4	23.8	24.4	22.4	19.0	(35)
MCDBR			23.1	26.2	27.6	28.1	29.1	27.2	24.8	26.1	27.9	29.2	26.9	22.4	(36)	
MCWBR			20.8	23.1	18.4	16.3	15.9	13.3	10.9	11.6	14.8	18.3	20.7	20.1	(37)	
5% WB		MCDBR	22.9	25.9	26.4	26.5	26.2	23.3	21.7	21.1	24.6	27.7	26.1	22.1	(38)	
		MCWBR	20.7	23.0	18.4	16.8	16.6	13.9	12.0	11.3	14.3	18.6	20.4	20.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.215	0.241	0.282	0.322	0.335	0.343	0.342	0.319	0.287	0.263	0.234	0.216	(40)	
(41)		taud	2.411	2.355	2.285	2.266	2.288	2.358	2.408	2.447	2.487	2.474	2.441	2.406	(41)	
(42)		Ebn at Noon	293	302	300	296	294	292	291	294	297	290	281	279	(42)	
(43)		Edh at Noon	21	28	35	40	40	38	36	33	29	24	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	633	975	1442	1872	2141	2102	1993	1805	1558	1077	680	516	(44)	
(45)		RadStd	42	46	56	76	142	122	107	96	74	56	38	28	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-5.13	N/A	N/A	-3.35	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-4.46	N/A	N/A	-1.48	-1.62	N/A	N/A	N/A	N/A

Nomenclature: See separate page